

Supervised Learning - Report

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1 Introduction

The sport of mixed-martial arts (MMA) is widely regarded as one of the most unpredictable- and dynamic sports one can spectate. A sudden knockout, freak injury or a single mistake giving way to a submission, it's a sport where anything, at any moment, can happen. An MMA bout is just as likely to be a controlled, one-sided domination as it is an explosive war of violence and unpredictability.

The UFC (Ultimate Fighting Championship) is currently the largest professional MMA organization with an extensive record of fights spanning across its 30+ years of existence. With such a long and lustrous history, finding a suitable dataset was not a difficult task. Kaggle, UFC Fight Data - 2026 became this chosen dataset.

With records dating back to March 2010 all the way to May of this year, it covers over 7000 fights and includes 57 features that describe each fighters stats., physical attributes and betting odds, and more. The set also includes a binary classifier on the winning fighter (whether the fighter in the red corner won or lost), which works as a great predictor feature.

Given this predictor feature we want to take note of the given baseline: As it stands, the red corner wins 58% of the time against the blue corner. With this benchmark, the main goal of this project then becomes two-fold:

- Develop a model that can outperform simple majority class baselines and accurately predict a UFC fight based purely on statistical differences between the fighters.
- See which attributes/features contribute the most to a outcome and by how much of a margin.

2 Process

Describe the process for finding the best model and justifying all decision. If you used other papers/sources for your decisions you need to reference them here.

After having picked a candidate dataset, we needed to verify its quality and usability for a potential model. For this part of the process we conducted some simple, but effective EDA that would answer that. We first checked for any obvious outliers and/or missing data values, of which there were few and far between. Next

2.1 Data Cleaning

After having conducted some initial pruning into the dataset and verified its usability we proceeded to a more extensive data cleaning.

The `_return` Columns The values of the `red_return` and `blue_return` columns reflect the payout a \$ 1 bet would return. Including them would cause target leakage since they tell the model exactly who won (loser's return being 0) and in turn lead to a 100% model prediction rate accuracy; something we obviously wanted to avoid.

Stances

We noticed in the `red_stance` and `blue_stance` columns that there were a few rows with faulty classifications and missing data ("Switch " with a trailing space and "Unknown"), that needed to be excluded. We also found one other classification called "Open Stance", which describes an opposite stance matchup, which was only found in the `blue_stance` column. Since we already have the three main stances ("Orthodox", "Southpaw" and "Switch") it becomes a redundant categorical value and only creates conflict.

Catch Weight

In the column `weight_class` there we found the value "Catch Weight" which describes fights where one or both fighters missed their target weight and therefore fought outside one of the 13 regulatory weight classes. This is by definition an outlier to the standard procedure of an MMA bout and should therefore be excluded.

Outliers in `_diff` Columns

In the `rounds_diff` column we found one extreme outlier where the value was ≥ 440 . We excluded this due to it being an impossible data value when looking at either fighters records (i.e. total fights).

In the `reach_diff` column there is a single outlier where the reach difference was ≥ 186 caused by one of the fighters reach being measured 0 cm. This is

an obvious error caused by a missing data value so we naturally excluded it.

3 Results

Report on the performance for the different models and hyperparameters. Essentially, take the numbers you got as output of your program and put them into a nice form (plots, tables, ...) that makes them easier to interpret and shows, for example, how the choice of values for different hyperparameters influences the performance.

4 Conclusion

Interpret the results and compare with results in the literature (if possible). What can be said about the performance of different models on the problem? How important is the choice of the hyperparameters for the performance of the different models?

5 Future Work

Suggest how your results could be improved. What did you learn from the results you got and what would you do differently now that you have learned this?